OXFORD AQA

INTERNATIONAL QUALIFICATIONS

Please write clearly in block capitals.

Centre number						Candidate number					

Surname ____________________

Forename(s) ____________________

Candidate signature ____________________

I declare this is my own work.

INTERNATIONAL GCSE
PHYSICS
Paper 1

Wednesday 8 May 2024 07:00 GMT Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:
- a pencil and a ruler
- a scientific calculator
- a protractor
- the Physics Equations Sheet (enclosed).

Instructions
- Use black ink or black ball-point pen.
- Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions in the spaces provided.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.
- In all calculations, show clearly how you worked out your answer.

Information
- The maximum mark for this paper is 90.
- The marks for questions are shown in brackets.
- You are expected to use a calculator where appropriate.

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
TOTAL	

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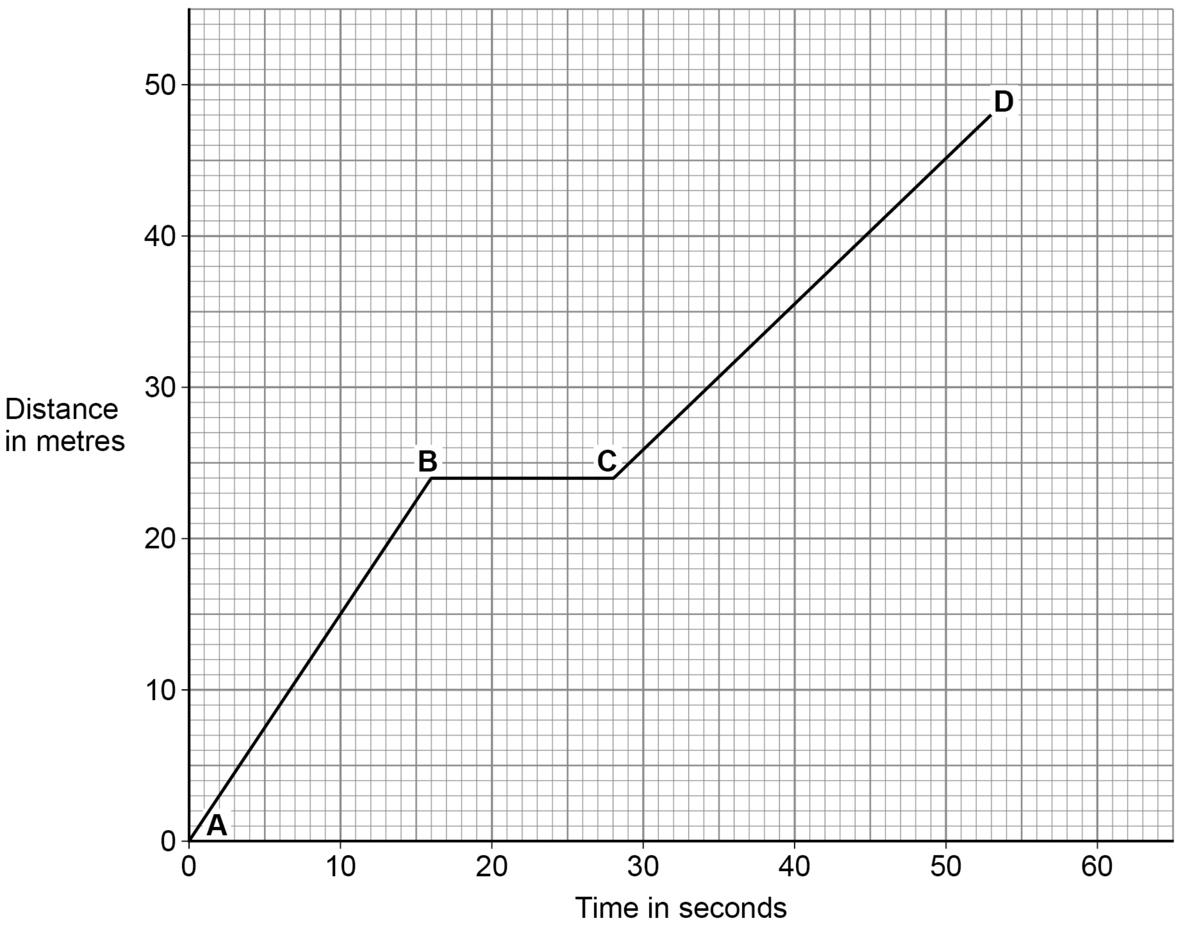
Answer **all** questions in the spaces provided.

0 1 **Figure 1** shows a distance-time graph for a child's walk.

Figure 1



0 1 . 1 Which of the following describes the motion of the child between **B** and **C** in **Figure 1**?

[1 mark]

Tick (✓) **one** box.

Constant acceleration ☐

Constant deceleration ☐

Constant speed ☐

Stationary ☐

0 1 . 2 Which of the following describes the motion of the child between **C** and **D** in **Figure 1**?

[1 mark]

Tick (✓) **one** box.

Constant acceleration ☐

Constant deceleration ☐

Constant speed ☐

Stationary ☐

Question 1 continues on the next page

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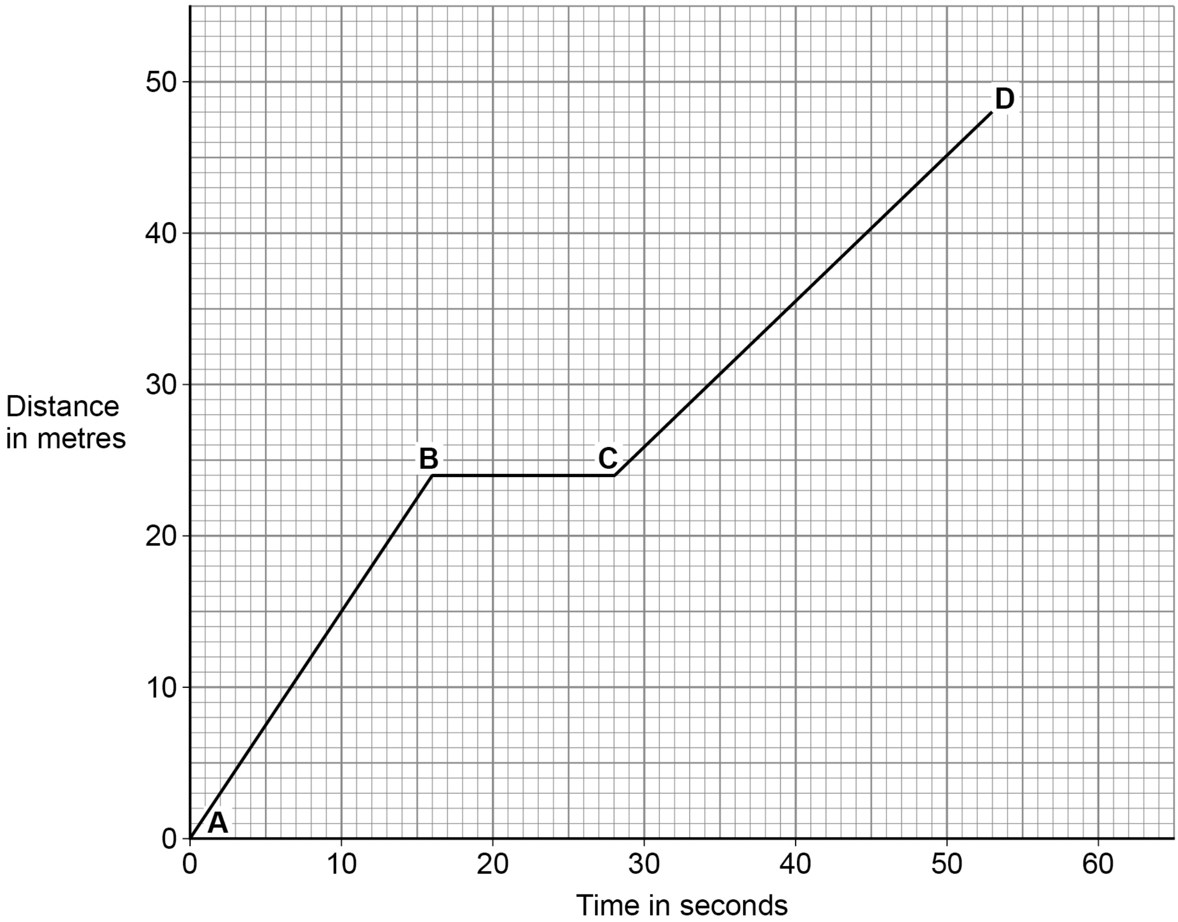
Figure 1 is repeated below.

Figure 1



0 1 . 3 What was the time taken between **C** and **D** in **Figure 1**?

[1 mark]

Tick (✓) **one** box.

12 seconds ☐

16 seconds ☐

25 seconds ☐

53 seconds ☐

Do not write outside the box

0 1 . 4 Calculate the speed of the child during the first 10 seconds.

Use **Figure 1** and the Physics Equations Sheet.

[3 marks]

Speed = ________________ m/s

0 1 . 5 How would the line **AB** in **Figure 1** be different if the child had walked faster?

[1 mark]

7

Turn over for the next question

Do not write outside the box

0 2 Sound is a longitudinal wave.

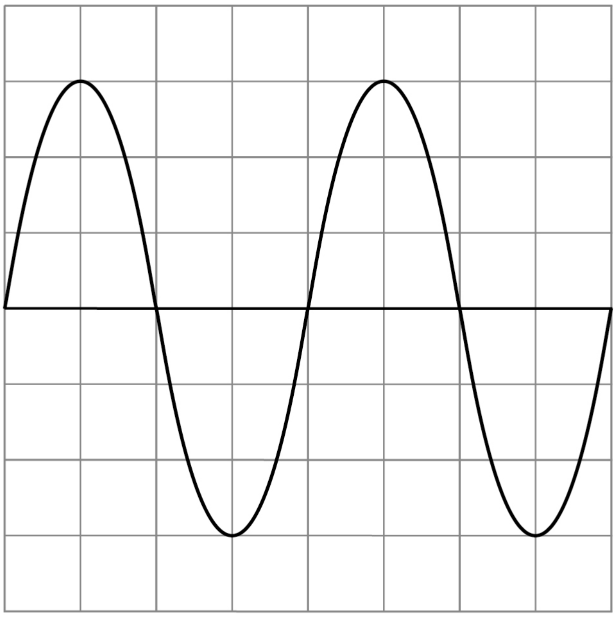
0 2 . 1 What is meant by a 'longitudinal wave'?

[1 mark]

Figure 2 shows a wave on an oscilloscope. The wave represents a sound wave.

Each square on the *x*-axis represents 0.001 seconds.

Figure 2



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0 2 . 2 Determine the period of the sound wave.

Use **Figure 2**.

[1 mark]

Tick (✓) **one** box.

0.001 s ☐

0.002 s ☐

0.004 s ☐

0.006 s ☐

0 2 . 3 Determine the frequency of the sound wave.

Use your answer from Question **02.2** and the equation:

$$\text{frequency} = \frac{1}{\text{period}}$$

[2 marks]

____

____

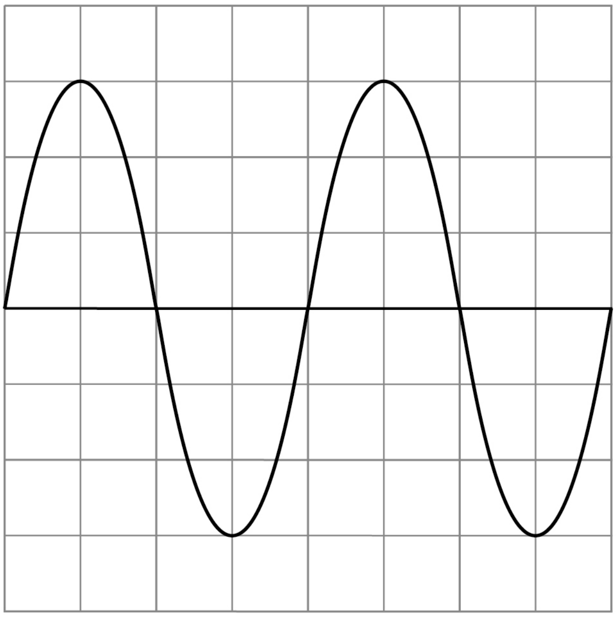
____

Frequency = ____________ Hz

Question 2 continues on the next page

0 2 . 4 **Figure 3** shows the same sound wave on the oscilloscope.

Figure 3



Draw a sound wave with the same frequency but a smaller amplitude on **Figure 3**.

[2 marks]

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Figure 4 shows a shark.

Figure 4

Sharks can detect low frequency sound waves from other fish in the sea.

0 2 . 5 A shark detects a sound wave from a fish.

The frequency of the sound wave is 20 Hz.

Sound waves travel at a speed of 1500 m/s in water.

Calculate the wavelength of the sound wave.

Use the Physics Equations Sheet.

[3 marks]

Wavelength = ____________________ m

Turn over ►

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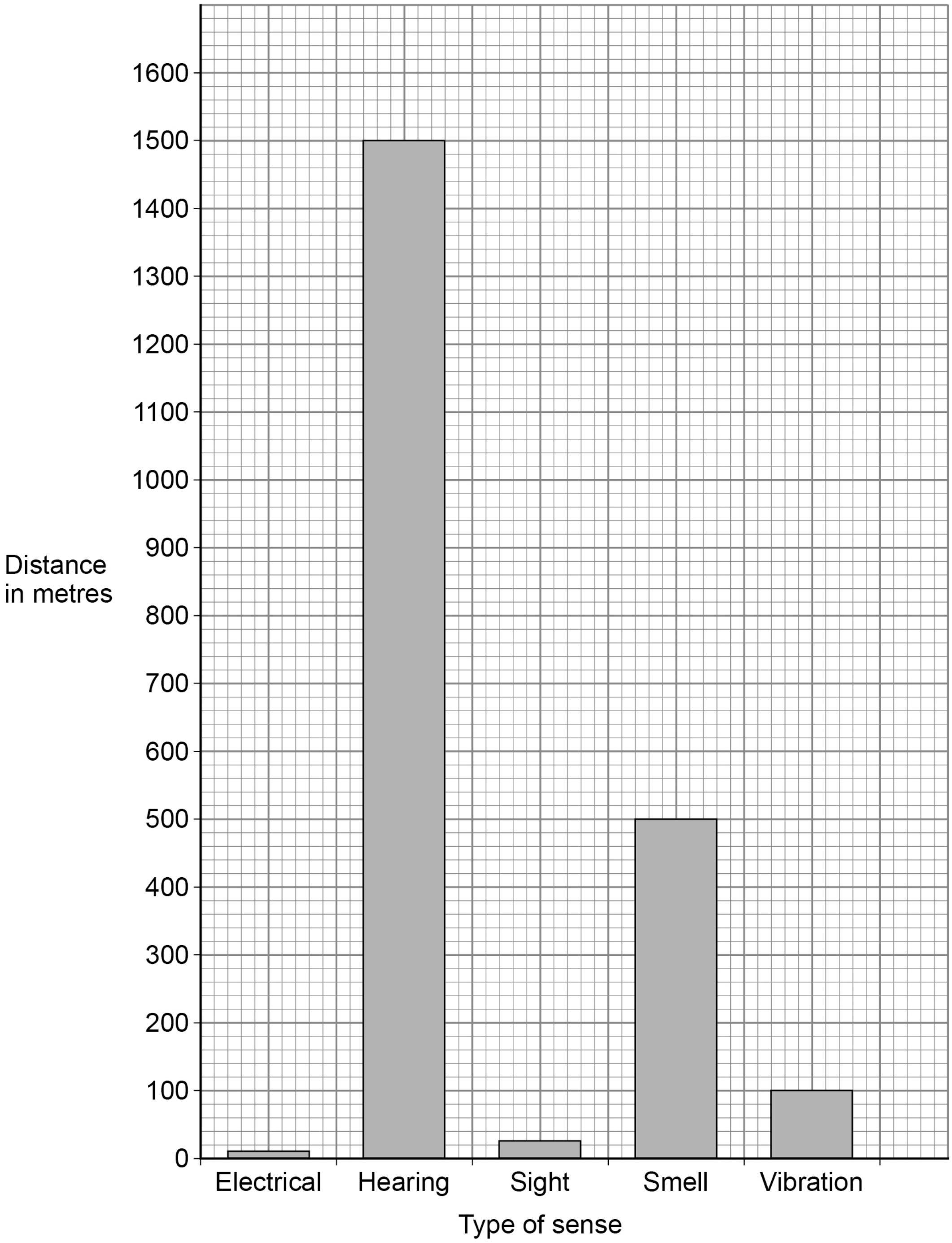
Sharks can use the senses of hearing, sight and smell to detect fish.

They also have electrical sensors and vibration sensors that they can use to detect fish.

Figure 5 shows the maximum distance at which the shark can detect fish using different senses.

Figure 5



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0 2 . 6 What is the maximum distance from which a shark can detect a fish?

Use **Figure 5**.

[1 mark]

Maximum distance = ____________________ m

0 2 . 7 Which senses could a shark use to detect a fish that was 400 metres away?

Use **Figure 5**.

[1 mark]

____

____

11

Turn over for the next question

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0 3 A student investigated how the length of a wire affected the resistance of the wire.

The student connected the wire into an electrical circuit containing a cell.

0 3 . 1 What is the independent variable in the investigation?

[1 mark]

Tick (✓) **one** box.

The length of the wire ☐

The resistance of the wire ☐

The temperature of the wire ☐

The type of metal the wire is made from ☐

0 3 . 2 What is the dependent variable in the investigation?

[1 mark]

Tick (✓) **one** box.

The length of the wire ☐

The resistance of the wire ☐

The temperature of the wire ☐

The type of metal the wire is made from ☐

Do not write outside the box

0 3 . 3 The student measured the current in the wire and the potential difference across the wire.

Draw **one** line from each quantity measured to the correct piece of equipment used.

[2 marks]

Quantity	**Equipment**
	Ammeter
Current	Joulemeter
	Metre rule
Potential difference	Power pack
	Voltmeter

Question 3 continues on the next page

The student used the measurement of current and potential difference to calculate the resistance of the wire.

The experiment was repeated for different lengths of wire.

Table 1 shows the results of the investigation.

Table 1

Length of wire in metres	**Resistance in ohms**
0.30	17
0.40	23
0.50	29
0.60	35
0.70	42
0.80	47
0.90	54

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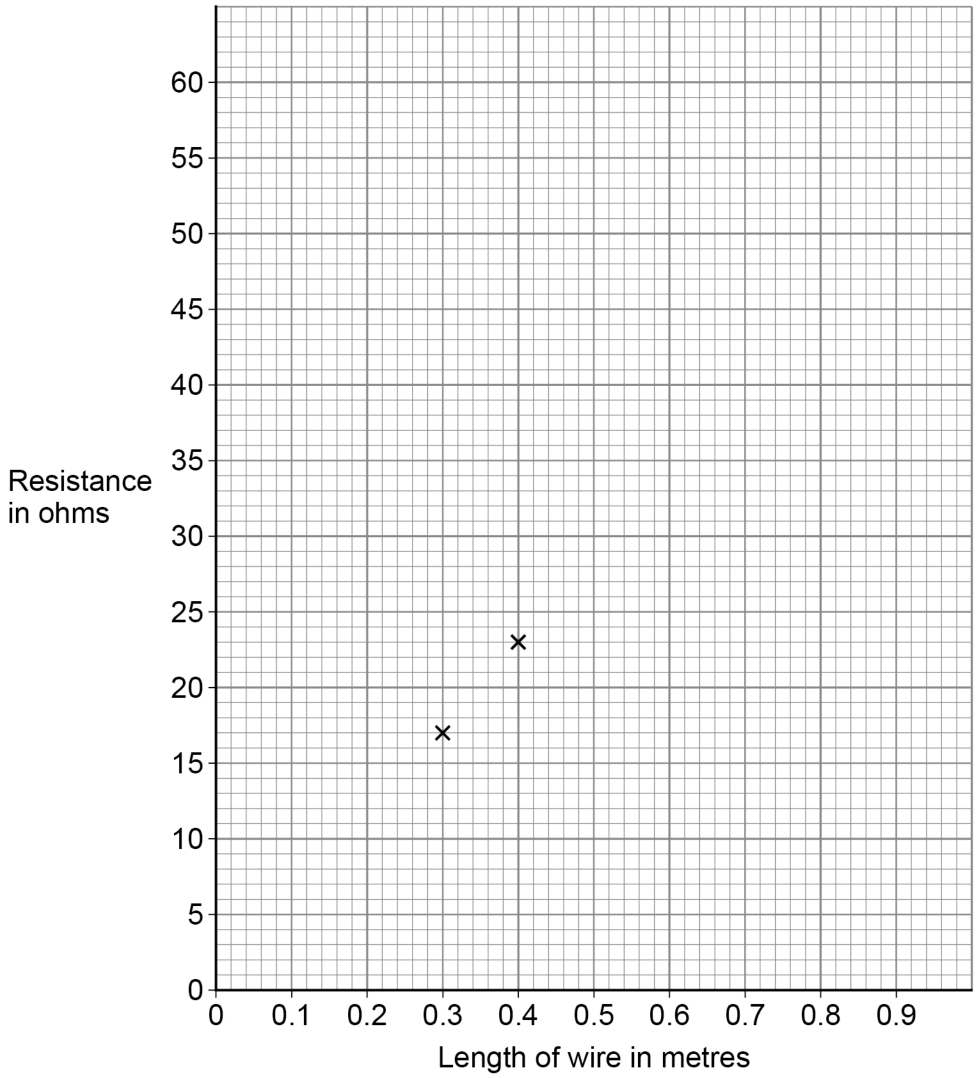
Figure 6 shows a graph of the results in **Table 1**.

Figure 6



0 3 . 4 Complete **Figure 6**.

You should:

- plot the remaining data from **Table 1**
- draw a line of best fit.

[3 marks]

Question 3 continues on the next page

Do not write outside the box

0 3 . 5 Describe the relationship between the length of the wire and the resistance of the wire.

[2 marks]

0 3 . 6 When a longer length of the wire was used the current in the wire was 0.20 A.

The potential difference across the wire was 12 V.

Calculate the resistance of the longer length of wire.

Use the Physics Equations Sheet.

[3 marks]

Resistance = ____________________ Ω

Do not write outside the box

0 3 . 7 The student kept the current in the wire at a small value.

Explain why a large current would affect the accuracy of the results.

Refer to electrons and ions in your answer.

[3 marks]

Question 3 continues on the next page

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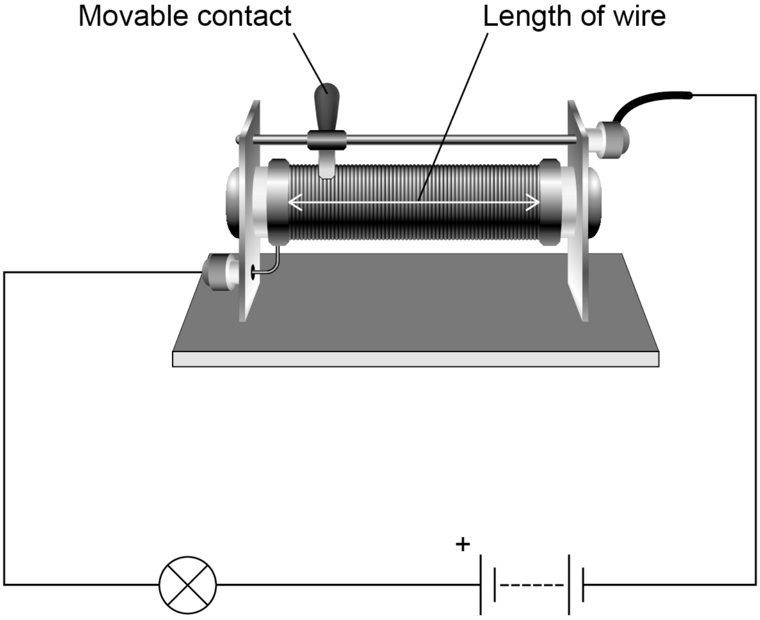
0 3 . 8 A variable resistor is connected in series with a battery and lamp.

Figure 7 shows the variable resistor in an electrical circuit.

Figure 7



The variable resistor has a moveable contact that is used to change the length of wire in the circuit.

Explain how the variable resistor can be used to make the lamp **less** bright.

[2 marks]

17

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Turn over for the next question

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0 4 The isotope polonium-210 is radioactive.

0 4 . 1 There are many different isotopes of polonium.

Describe what isotopes are.

[2 marks]

____

____

____

____

0 4 . 2 When a nucleus of polonium-210 decays it emits an alpha particle.

Give the structure of an alpha particle.

[1 mark]

____

____

0 4 . 3 Alpha particles are charged.

Which **two** of the following are also properties of alpha particles?

[2 marks]

Tick (✓) **two** boxes.

Can pass through aluminium	☐
Can travel a few metres through air	☐
Most ionising type of radiation	☐
Move at the speed of light	☐
Stopped by paper	☐

Question 4 continues on the next page

Figure 8 shows paper moving over rollers in a paper-making factory.

When the paper moves over the rollers it becomes charged with static electricity.

Radiation from polonium-210 can be used to remove static electricity from paper.

Figure 8

Do not write outside the box

0 4 . 4 Explain how the paper becomes negatively charged with static electricity as it moves over the rollers.

[3 marks]

The radiation from the polonium-210 ionises the air near the paper on the rollers.

0 4 . 5 What is an ion?

[1 mark]

0 4 . 6 The ions reduce the static electricity.

Explain how the ions can reduce the static electricity on the negatively charged paper.

[2 marks]

11

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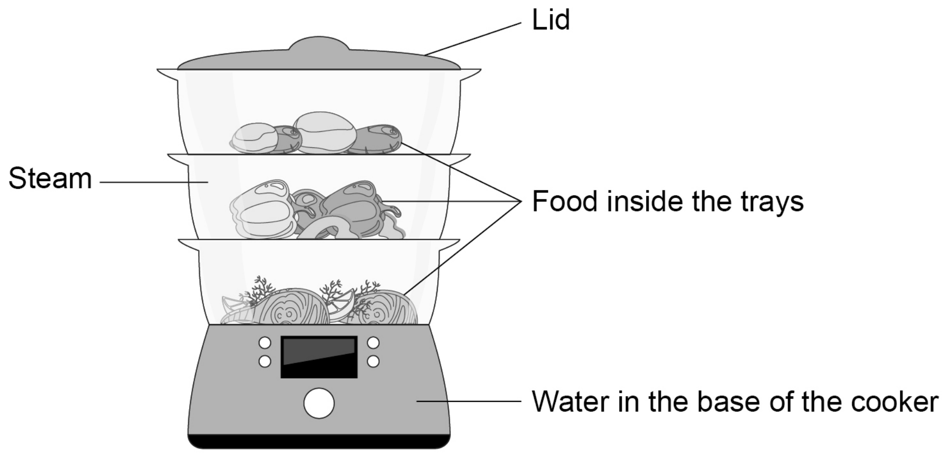
0 5 **Figure 9** shows an electric cooker that uses steam to cook food.

Figure 9



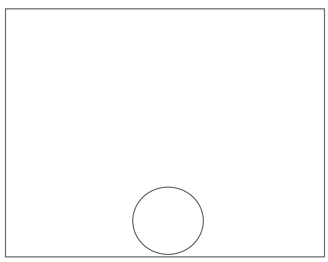
Water is poured into the base of the cooker, where it is heated to boiling point.

The steam rises and fills the cooker. The steam cooks the food.

0 5 . 1 Draw the arrangement of particles in liquid water.

One particle has already been drawn.

[1 mark]



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0 5 . 2 The water poured into the cooker has a temperature of 15 °C.

The temperature of the water is increased to 100 °C.

The water poured into the cooker has a mass of 0.60 kg.

specific heat capacity of water = 4200 J/kg °C.

Calculate the energy transferred to the water to raise its temperature to 100 °C.

Use the Physics Equations Sheet.

[3 marks]

Energy transferred = ____________________ J

0 5 . 3 When the water is at its boiling point, some of the water changes state.

Describe how the arrangement of the water particles changes as the water changes state.

[1 mark]

Question 5 continues on the next page

Do not write outside the box

0 5 . 4 The mass of water that changes state is 175 g.

specific latent heat of vaporisation of water = 2 300 000 J/kg

Calculate the energy transferred to the water to change state.

Give your answer to 3 significant figures.

[3 marks]

Energy transferred (3 significant figures) = ____________ J

0 5 . 5 The cooker is turned off and the lid is removed.

Some of the water left in the cooker evaporates.

Explain why evaporation causes the temperature of the water in the cooker to decrease.

[4 marks]

12

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0 6 Resultant forces affect the motion of cars.

0 6 . 1 What is meant by 'resultant force'?

[1 mark]

____

____

0 6 . 2 A car is stationary. What is the resultant force acting on the car?

[1 mark]

____

____

Question 6 continues on the next page

Do not write outside the box

0 6 . 3 A resultant force of 2 700 N acts on the car.

The mass of the car is 900 kg.

Calculate the acceleration of the car.

Use the Physics Equations Sheet.

Give the unit.

[4 marks]

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Acceleration = ____________________ Unit __________

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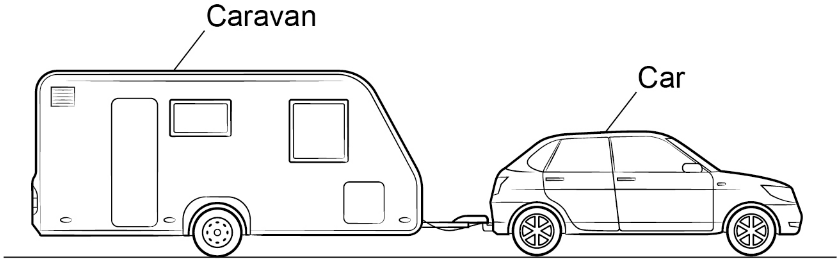
Figure 10 shows the car pulling a caravan along a road.

Figure 10



0 6 . 4 The car in **Figure 10** exerts a force on the caravan and the caravan exerts a force on the car.

Explain the relationship between the magnitude and direction of these two forces.

[3 marks]

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Question 6 continues on the next page

Do not write outside the box

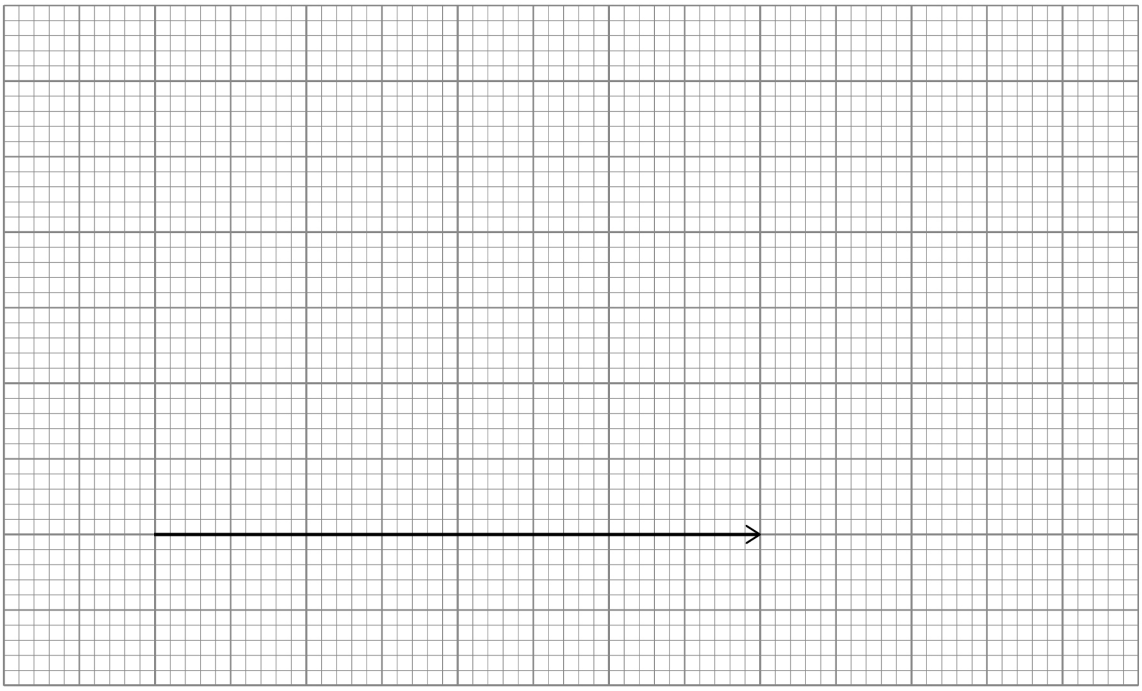
0 6 . 5 At a different stage of the journey the car exerts a forward force of 800 N on the caravan.

A sudden gust of wind exerts a force of 500 N on the caravan at an angle of 60° to the 800 N force.

The 800 N force has been drawn to scale for you.

Determine the magnitude of the resultant of these two forces by drawing a vector diagram to scale.

[3 marks]



Resultant force = ____________________ N

12

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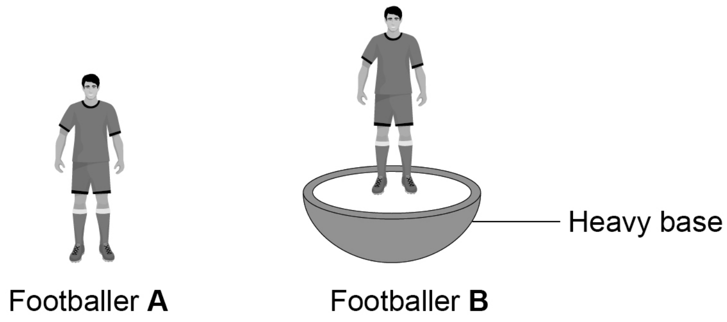
0 7 Table-top football is a game that uses plastic footballers.

Figure 11 shows two plastic footballers.

Figure 11



0 7 . 1 Footballer **A** has no base. Footballer **B** is on a heavy base.

Give **two** reasons why footballer **B** is more stable than footballer **A**.

[2 marks]

1 ____

____

2 ____

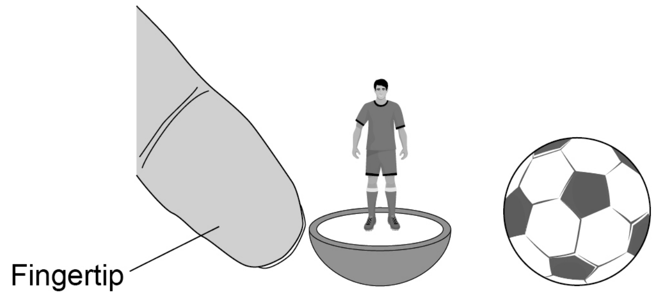
____

Question 7 continues on the next page

Do not write outside the box

Figure 12 shows footballer **B** being pushed towards a stationary ball.

Figure 12



0 7 . 2 When footballer **B** collides with the stationary ball, the ball accelerates.

Explain why.

Use conservation of momentum in your answer.

[4 marks]

Do not write outside the box

0 7 . 3 Footballer **B** applies a force of 0.0090 N to the stationary ball.

Footballer **B** is in contact with the ball for a time of 0.12 s.

The mass of the ball is 0.0015 kg.

Calculate the speed of the ball after footballer **B** has been in contact with the ball.

[5 marks]

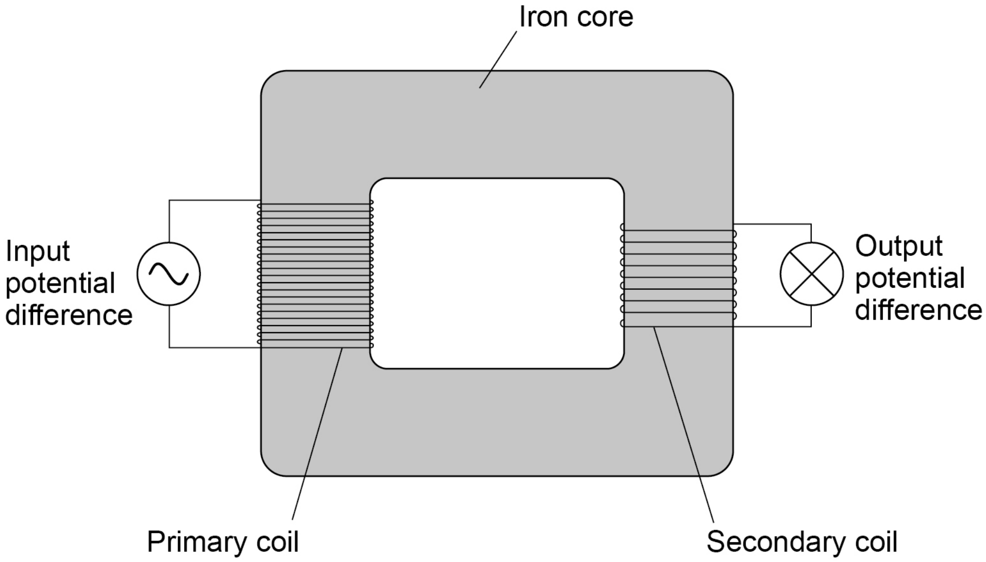
Speed = ____________________ m/s

11

Turn over for the next question

0 8 **Figure 13** shows a basic transformer.

Figure 13



0 8 . 1 What type of basic transformer is shown in **Figure 13**?

[1 mark]

____

____

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0 8 . 2 The transformer is 100% efficient.

The current in the secondary coil is different from the current in the primary coil.

Explain why.

[4 marks]

Question 8 continues on the next page

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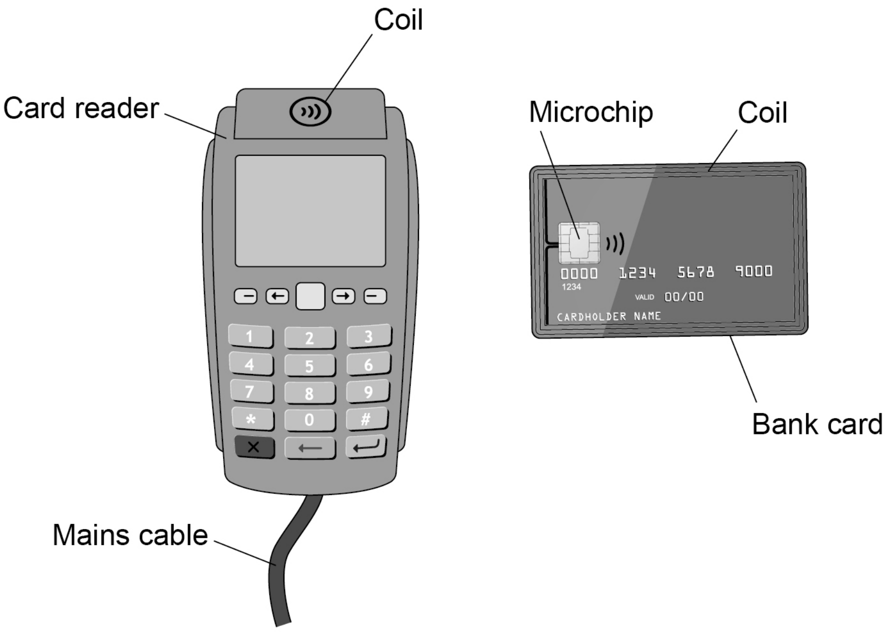
Contactless bank cards do not have their own power source.

Figure 14 shows a card reader and contactless bank card.

Figure 14



The card reader contains a coil connected to the mains supply.

The bank card has a coil connected to the circuit of the microchip.

The coil in the bank card acts like the secondary coil in a transformer.

0 8 . 3 There is a current in the microchip when the bank card is placed near the card reader.

Explain why.

[3 marks]

Do not write outside the box

0 8 . 4 Suggest **one** advantage of a bank card that does not require its own power source compared to a bank card that does.

[1 mark]

9

END OF QUESTIONS

There are no questions printed on this page

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Question number	**Additional page, if required. Write the question numbers in the left-hand margin.**